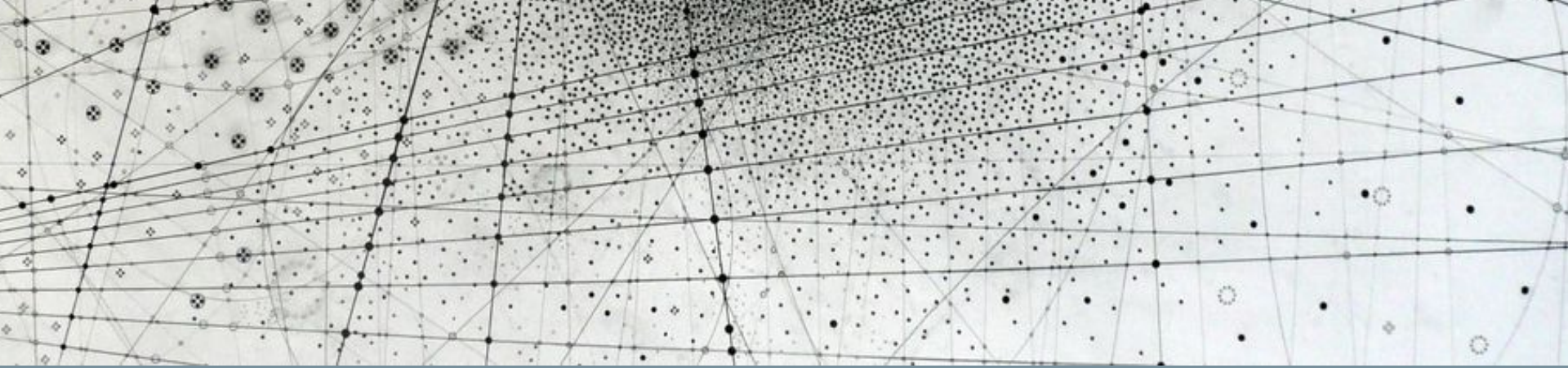




Research Methods in Machine Learning & Physics

Coming up...

Due Wednesday: Pset #2 & peer review of Project #1

Upcoming lecture topics:

LLMs, Self-Supervised Learning, Multimodal Models, & Foundation Models

Next project topic: GNNs & astrophysics data

Coming up...

Monday 11/24: Giving a good academic talk

Wednesday 11/26: Off for Thanksgiving

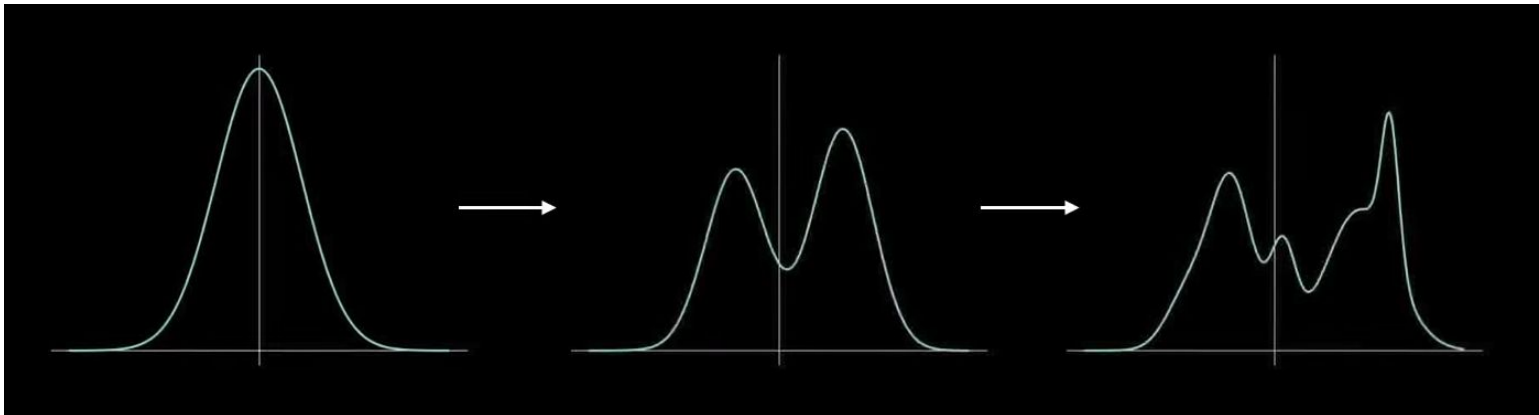
Monday 12/1 & Wednesday 12/3: No lectures (NeurIPS week!)
→ Instead, we'll schedule one-on-one project consultations via Zoom

Monday 12/8 & Wednesday 12/10: Final in-class presentations

Quick recap

Quick Recap: Flow-based methods

Key idea: learn a mapping from a simple distribution to a complex one



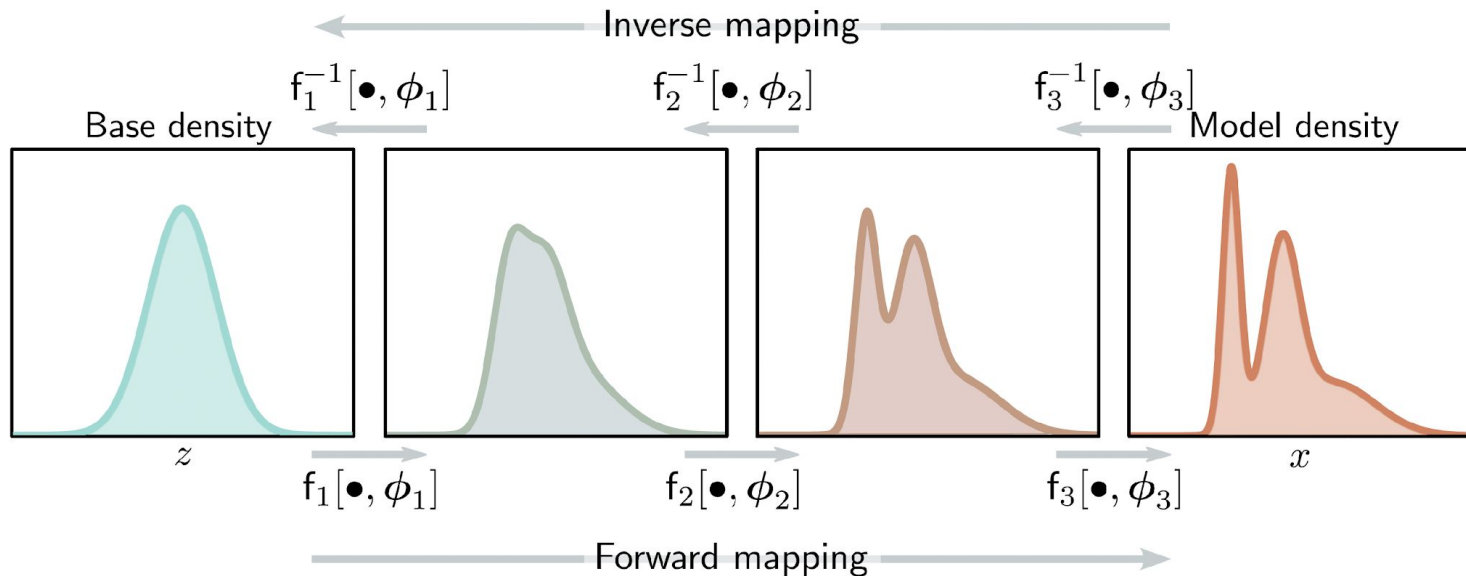
Quick Recap: Flow-based methods



Quick Recap: Flow-based methods

Want to calculate the exact likelihood distribution? → **Normalizing Flow**

$$\mathbf{x} = \mathbf{f}[\mathbf{z}, \phi] = \mathbf{f}_K \left[\mathbf{f}_{K-1} \left[\dots \mathbf{f}_2 \left[\mathbf{f}_1[\mathbf{z}, \phi_1], \phi_2 \right], \dots, \phi_{K-1} \right], \phi_K \right]$$

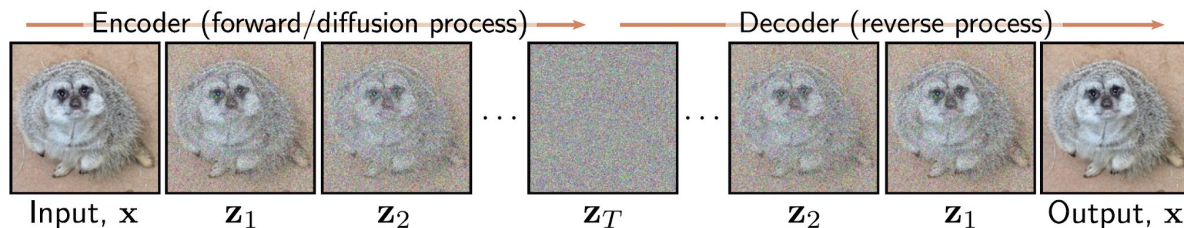


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Want to get the highest-quality generated outputs? → **Diffusion Model**

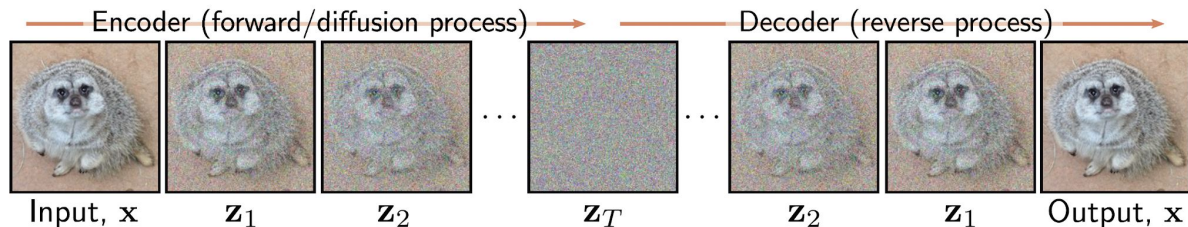


Quick Recap: Flow-based methods

Want to calculate the exact likelihood distribution? → **Normalizing Flow**

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Want to get the highest-quality generated outputs? → **Diffusion Model**



Want something in between? → **Flow-matching**

Transformers, Part I

Attention Is All You Need

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Anatomy of a Transformer layer

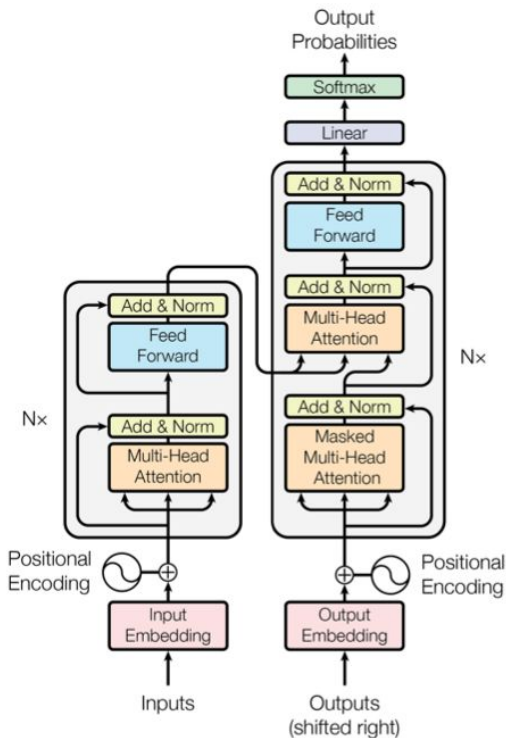
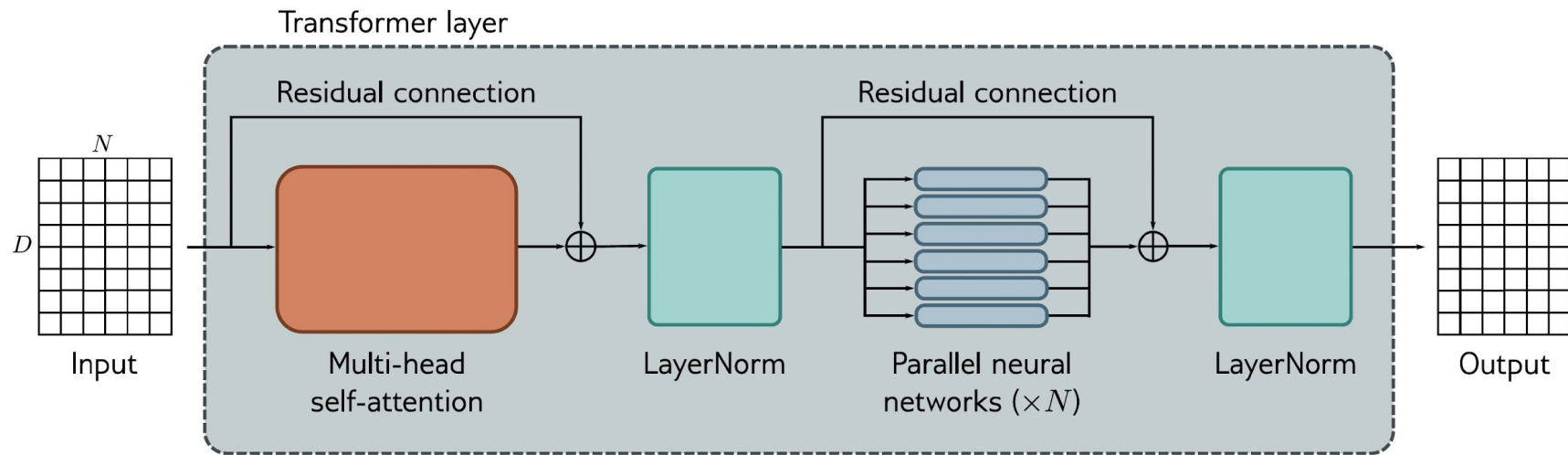


Figure 1: The Transformer - model architecture.

Anatomy of a Transformer layer



Learning a richer
representation of your
input data

Asking useful questions
from your embeddings

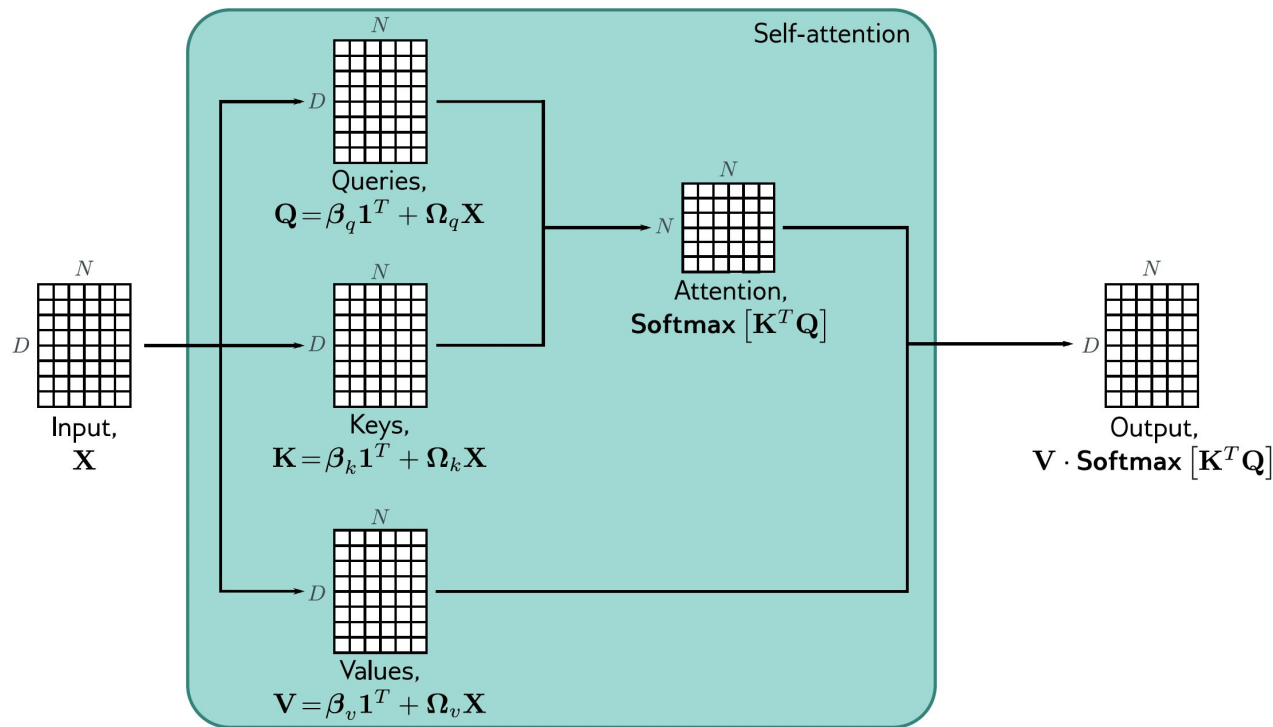


Inside an
LLM

Inside the self-attention block...

- The goal of self-attention is to find important relationships between pairs of words → **dot product**
- We want to do this in a **computationally-efficient** manner
 - Easily scalable
- We want it to be able to process **variable-length inputs**
- And we also want this to reflect **permutation equivariance**

Inside the self-attention block...



Inside the self-attention block...

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Inside the self-attention block...

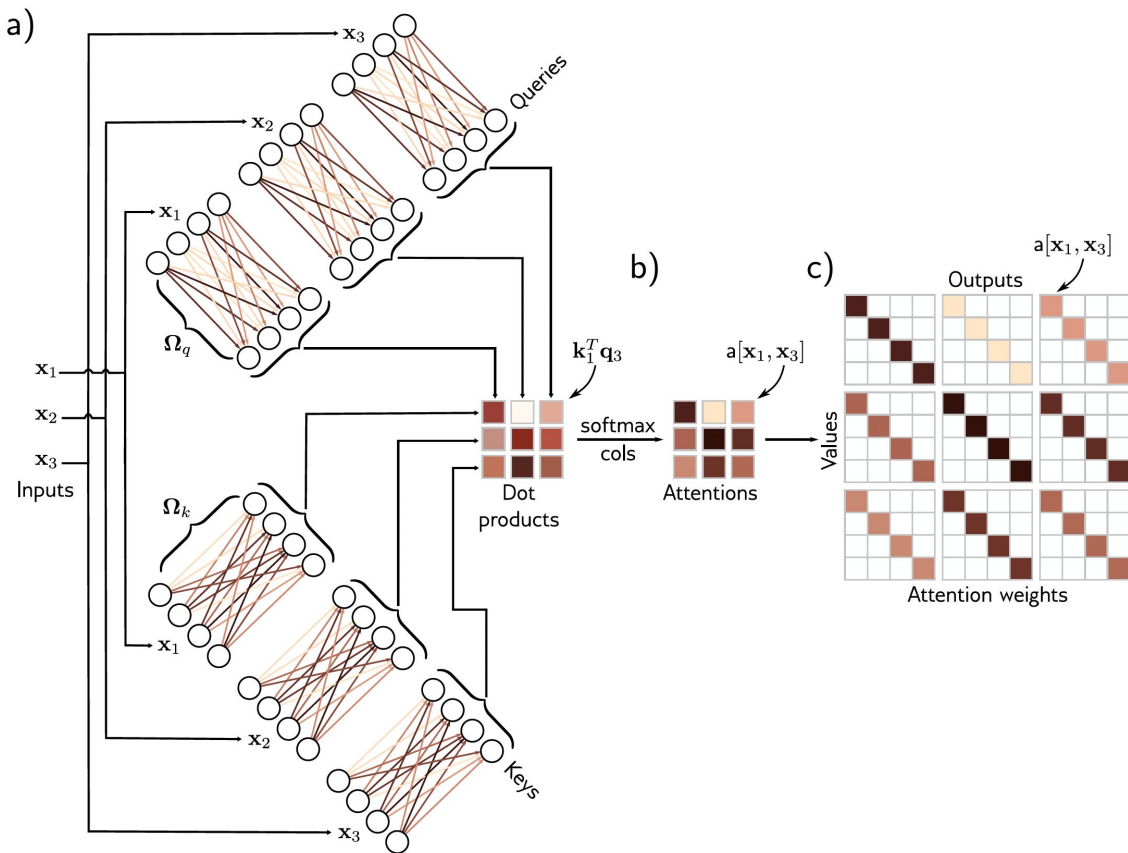
$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{Q}} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{Q} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{K}} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{K} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

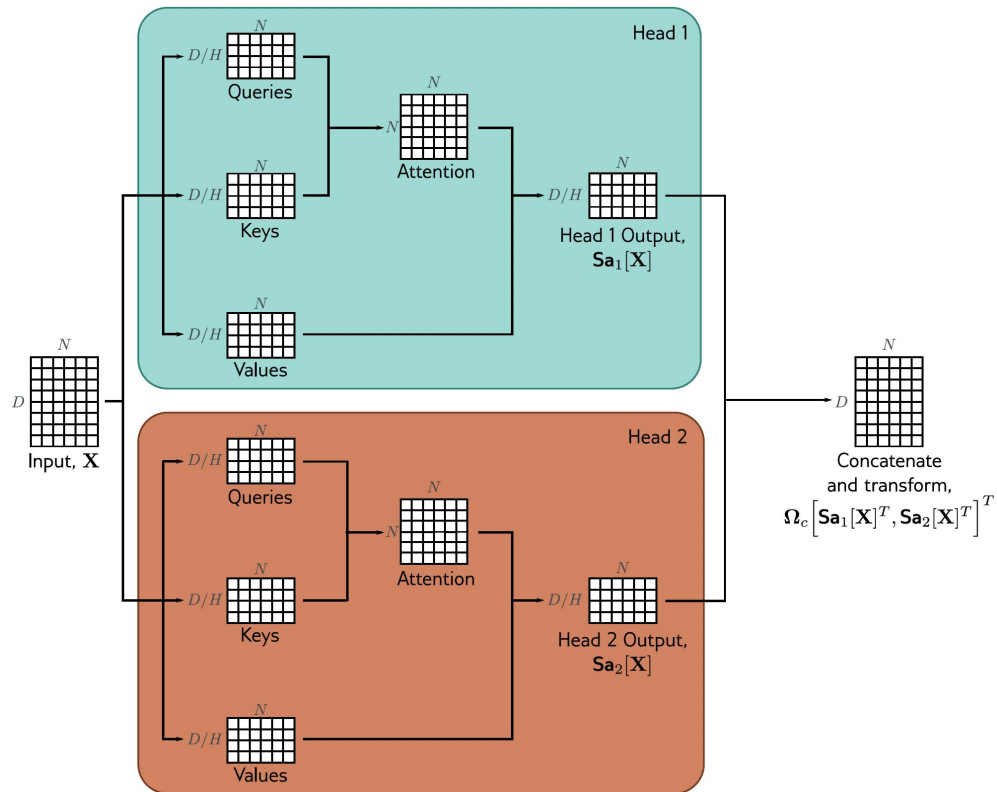
$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{V}} \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{V} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

$$\begin{aligned} & \text{softmax} \left(\frac{\begin{matrix} \text{Q} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{K}^{\text{T}} \\ \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \end{matrix}}{\sqrt{d_k}} \right) \begin{matrix} \text{V} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix} \\ &= \begin{matrix} \text{Z} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix} \end{aligned}$$

Inside the self-attention block...



Multi-head attention



Recommended resources

- Writing:
 - Understanding Deep Learning, Chapter 12
 - Attention is All You Need (original paper)
 - The Illustrated Transformer
 - Transformers from Scratch
- Code:
 - Visualizing self-attention with Tensor2Tensor
- Video lectures:
 - <https://youtu.be/wjZofJX0v4M>
 - <https://youtu.be/eMlx5fFNoYc>
 - <https://youtu.be/phOc25QfNS0>

In-class work

The Annotated Transformer:

<http://nlp.seas.harvard.edu/annotated-transformer/>

→ *Click through the link to the codebase / Colab notebook*